



Far transfer effects of executive working memory training on cognitive flexibility

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Accepted: 14 June 2022

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Abstract

The near and far transfer effects of Working Memory (WM) training have yielded discrepant findings while there is no literature on the effects of Executive Working Memory (EWM) training. The present study aimed to investigate the far transfer effects of EWM on cognitive flexibility. Community participants ($n = 144$) were allocated into a fully-, a partially-trained or a control group. At a baseline assessment, all groups were administered the Wisconsin Card Sorting test and Letter-Number Sequencing (LNS). Following this and for six consecutive days, the fully-trained group were administered the complete LNS, the partially-trained group were administered the LNS up to the three-digit strings while the control group had no involvement in the study. Upon completion of training, all groups were administered the Intra/Extra-Dimensional Shift task (ID/ED). The control group (a) made more errors and completed fewer stages compared with the fully-trained group and (b) had increased response latency and required more trials to complete the ID/ED compared to both the partially- and fully-trained groups. In the total sample, (a) the lack of EWM training was associated with more total errors, trials required to complete the ID/ED, extradimensional-shift errors and prolonged response latency and (b) extended short-term EWM training was associated with more completed stages in the ID/ED. The findings support the far transfer effects of EWM training on cognitive flexibility and are in agreement with studies on WM training. Although preliminary, they hold promise for the enhancement of early intervention programs in populations with cognitive impairments.

Keywords Executive working memory training · Cognitive flexibility · Far transfer effects · Neuroplasticity · Early intervention

Introduction

Working memory (WM) is a core cognitive function referring to the ability to briefly retain and recall a small amount of information (Aben et al., 2012) and was well acknowledged when (Baddeley & Hitch, 1974) proposed their respective model. Executive working memory (EWM) refers to the capacities involved in WM, with special emphasis on the manipulation of retained information prior to recall (Perry et al., 2001). The use of neuroimaging technology has allowed a more thorough investigation of the neural substrates of WM and EWM, revealing that they are both mediated by the activation and integrity of a prefrontal-temporal-parietal circuitry (Tronchin et al., 2020). Other researchers (Christophel et al., 2017, p. 120), thus, proposed that “WM is supported by a distributed network throughout the brain that gradually transforms sensory information towards an appropriate behavioural response, across a temporal delay”

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and it seems that this distributed neural network also supports EWM.

Initially, WM was widely acknowledged as having limited capacity (Baddeley, 2010). This view however has been reconsidered as studies revealed that WM capacity can be increased via training with WM tasks (Harrison et al., 2013). The phenomenon was named *near transfer* and as Green et al. (2019, p. 10) noted “*Near transfer is typically used to describe cases where training on one task produces benefits on tasks meant to tap the same core construct as the trained task using slightly different stimuli or setups*”; near transfer effects of WM training have been consistently observed in children (Nelwan et al., 2018), adults (Etherton et al., 2019) and older populations (Borella et al., 2017). In addition, studies have demonstrated that WM training can lead to improvements in other cognitive domains (e.g. Etherton et al., 2019; Jaeggi et al., 2014). This phenomenon was named *far transfer*, it refers to the improvement of cognitive functions that share common neural substrates with WM (Caeyenberghs et al., 2016) after WM training and as per Green et al. (2019, p. 10) it refers to “...*situations where the training and transfer tasks are not believed to tap the exact same core construct. In most cases, this means partial, but not complete, overlap between the training and transfer tasks...*”. The theories on the transfer effects of WM involve mechanisms of brain plasticity, enhancement of WM sub-processes that are further activated by the training task or acquisition of new skills that are implemented when performing untrained tasks (Gathercole et al., 2019). Although the near transfer effects of WM training have been well-documented, there is still an ongoing debate regarding the effectiveness of WM training on far transfer gains (Melby-Lervåg & Hulme, 2013). To the best of our knowledge, there are currently no studies examining the near and far transfer effects of EWM training.

A prominent association of WM and EWM is with cognitive flexibility that is the ability of “...*individuals to update their plans or expectations in response to new information, explore alternative strategies for solving problems, and generally adapt behaviors to changing environmental demand*” (Van Stockum & DeCaro, 2020, p.2). Cognitive flexibility is also mediated by a prefrontal-temporal-parietal neural circuitry (Dajani & Uddin, 2015) and shares a set of common underlying processes, such as effective switching between responses, maintaining at least two rules for effective performance and inhibitory control, with EWM. A fairly small set of studies have investigated the effectiveness of WM training on cognitive flexibility in specific populations. In detail, (a) researchers (Chiu et al., 2018) tested healthy older adults and reported that the WM training group (30-min training sessions, 3 /week for 8 weeks) had higher cognitive flexibility in the follow

up session (a period of 3–6 months after training); (b) researchers (Fraser & Cockcroft, 2020) examined a clinical population (HIV+ adolescents) who received adaptive WM training for 8 weeks resulting in improved verbal WM and transfer effects in attention, aspects of executive function, memory, language and fluid intelligence - these effects were maintained for at least 6 months following the intervention and (c) researchers (de Vries et al., 2015) applied an 8-week WM protocol in a sample of autistic children, which led to improvements in cognitive flexibility. However, other studies have yielded opposite findings. Specifically, they (Goghari & Lawlor-Savage, 2017) found no far transfer effects after 8-week training of WM or logic and planning on cognitive flexibility in healthy older adults above the age of 65. In another study (Maraver et al., 2016), the training of processes involved in WM (maintenance, updating and memory search processes) for six sessions distributed over 2 weeks, did not lead to far transfer effects in a task of proactive/reactive cognitive control, which could be considered an index of cognitive flexibility. Furthermore, cognitive training with the use of an adaptive procedure over 21 days (10 training sessions) on executive functions (including cognitive flexibility) did not reveal any near or far transfer effects, only some evidence of direct training (De Lillo et al., 2021). These differences in findings could be due to sample characteristics, for example, cognitive flexibility in children might not be “mature enough” (Buttelmann & Karbach, 2017) for training to exert its effects, ageing might have led to cognitive decline in older adults so severe that they cannot benefit from training due to decreased brain plasticity (Burke & Barnes, 2006), or other methodological issues of the studies such as differences in the duration and tasks of training, inclusion of student participants in both trained and untrained groups although education by itself is known to serve as a means of cognitive training (Ritchie & Tucker-Drob, 2018).

Although limited, the previously described studies highlight the significance of the effects of WM training on cognitive flexibility; what is missing from the literature, though, is the examination of the effects of the more complex EWM on the specified higher-order cognitive function. To this end and given the commonalities in the neural substrate underlying EWM and cognitive flexibility, the aim of the present study was to investigate the effects of short-term adaptive EWM training on cognitive flexibility in healthy adults. We hypothesized that a linear relationship would be revealed, such that participants engaging in a full-training protocol would present with superior performance in the cognitive flexibility task administered post-training, followed by those allocated to a partial-training protocol, followed by a control group that did not receive any form of EWM training.

Methods

Participants

One hundred and forty-four healthy community participants were recruited [age mean $\pm$ SD: 28.38 ± 12.35 , age range: 19–45 years; males (54.9%) /females (45.1%)]. Exclusion criteria included personal history of head trauma, medical or neurological conditions, current use of prescribed or recreational drugs and personal or family (up to second-degree) history of DSM-IV Axis I disorders. All participants were screened by a psychologist (V.S.) for the fulfillment of the aforementioned criteria. The study was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki and its later amendments and was approved by the Research Ethics Committee of the University of Crete (registration number: 10/24-04-2015). Following oral presentation of the study's aims and methods, all participants received a detailed information sheet and gave written informed consent before inclusion in the study.

One participant was excluded from the analyses as she was identified as an outlier in most neuropsychological measures. Therefore, the final sample consisted of 143 participants [age mean $\pm$ SD: 27.24 ± 6.26 , age range: 19–45 years; males (55.2%) /females (44.8%)].

Neuropsychological assessment

Raven's standard progressive matrices (Raven, 2003) The test assesses abstract reasoning. It consists of 5 sets of 12 abstract patterns, each with one missing piece. Participants were required to select the piece that best completed the pattern among a set of alternatives. Items within a set were of increasing difficulty. The outcome measure was the total number of correct selections.

Wisconsin Card Sorting Test (WCST) (Nelson, 1976) The test assesses cognitive flexibility (Gläscher et al., 2019); a computerized version was employed in the present study. Four cards with stimuli that differed in three dimensions (color, shape, number) and a target card were presented on the screen. Participants were required to match the target card with one of the stimulus cards according to a sorting principle (the sorting principle corresponded to one of the aforementioned dimensions) and feedback was provided after each match. In the first part of the task, the first sorting principle applied by the participant was accepted as correct. When 6 consecutive correct responses were given, the sorting principle changed, feedback was provided to the participants and they were required to find a new sorting

principle according to another dimension. In the next two matches, the two remaining sorting principles were selected. In the second part of the task the same sorting principles should be re-selected in the same order, as in the first part. The task was discontinued when the participant failed to identify the sorting principles or the target cards were exhausted. The outcome measures were the number of completed categories, number of unrelated matched cards (i.e. the target card had nothing in common with the selected card), perseverative errors, non-perseverative errors and total number of errors.

Letter-Number Sequencing (LNS) (Stogiannidou, 2014) The test assesses EWM as it requires both temporary storage of information and their manipulation prior to recall. In this study, we administered the Greek version of the task (Stogiannidou, 2014). Strings of intermingled letters and numbers were read to the participants (strings were of increasing difficulty) who were required to recall the strings after reordering of the information (i.e. placing first all numbers in numerical order and then all letters in alphabetical order). The test consists of eight blocks with three trials each. The outcome measure was the total number of correct strings.

Intra-Extra Dimensional Shift (ID/EDS) (Robbins et al., 1998) The test is part of the Cambridge Neuropsychological Test Automated Battery, it assesses cognitive flexibility and is administered using a computer with a touch-sensitive screen (Robbins et al., 1998). At each stage of the task, participants were presented with stimuli consisting of two dimensions (shape and line) and they had to acquire a stimulus-reward association: they had to use feedback to work out a rule that was given after each correct response. After six correct responses, the rule changed. The task initially involved simple shape discrimination and reversal and subjects proceeded to tests of distraction where the second dimension (line) was introduced, first adjacent to the shapes and then superimposed on them. Once training in simple discrimination and reversal stages was complete, the intra and extra-dimensional shift stages took place. In the first stage (intra-dimensional shift) the rule did not change, despite the presentation of new stimuli. When discrimination had been established to these new stimuli, a second shift occurred (extra-dimensional shift) in which the irrelevant dimension (line) became relevant. Outcome measures were total errors, EDS errors (errors made in the extra-dimensional stage of the task), Pre-ED errors (number of errors made in the intra-dimensional shift), completed stage errors (the number of errors made on stages successfully completed), completed stage trials (the number of trials undertaken on all successfully completed stages) completed stages and latency (in msec).

Procedure

All participants took part in a baseline assessment session, in which they were administered the RPM, WCST and LNS. Subsequently, they were allocated to three groups: fully-trained group, partially-trained group and control group. The three groups were matched for demographic characteristics and baseline neuropsychological task performance.

According to the training protocol, after completion of the baseline assessment session participants in the fully-trained group were administered the LNS up to the last string (21 trials in total) for six consecutive days (one training session/day), participants in the partially-trained group were administered the LNS up to the strings with three digits (6 trials in total) for six consecutive days (one training session/day) and participants in the control group had no involvement in the study for six days. Although the training period was shorter compared to the literature on working memory training (for a meta-analysis see, Au et al., 2015) we chose this length of training due to (a) high drop-out rates of our healthy participants during a pilot study preceding the present one, who reported that the procedure “became quite tedious” after a week of training and (b) our interest in examining the effects of shorter-than-usual training periods on transfer gains (both points are raised in the meta-analysis by Au et al., 2015). The day after completion of the training period (i.e. on day 7 after the baseline assessment), all participants were administered the ID/EDS.

Green et al., (2019) have proposed a set of best-practice methodological standards that could be applied in cognitive-training/cognitive-enhancement studies. According to this set of criteria, the present study could be identified as a *feasibility study* as it aims to examine the viability of the described training method before applying the procedures to clinical/high-risk populations (a potential future effectiveness study). In this respect, (a) the relatively small sample size is acceptable, although the inclusion of healthy participants does not meet the criterion of selecting participants representative of the populations for the studies to follow; (b) the lack of including an active control group does not limit the value of the study as the inclusion of a passive control group serves its aims; (c) matched groups of participants were included in order to avoid the effects of potential confounders and (d) outcome variables can be effectively measured in future effectiveness studies.

Statistical analyses

All statistical analyses were performed using SPSS version 25 (SPSS Inc., Chicago, Illinois). Between-group differences in demographic variables and neuropsychological task performance were examined either with parametric or non-parametric analyses according to normality of the distribution as

examined with the Kolmogorov–Smirnov test. Statistically significant differences between the groups were followed up with Bonferroni post-hoc test or Mann-Whitney tests, as appropriate. Between-group differences in sex were examined with Pearson’s chi-square.

The association between baseline neuropsychological function, EWM training and cognitive flexibility following training was examined with a series of stepwise regressions. For the purposes of these analyses, we calculated a new variable named baseline cognitive flexibility by transforming outcome measures of the WCST (except total errors) into z-scores and then calculating the mean for every participant. Thus, in the regression analyses the dependent variables were the ID/ED outcome measures (included one by one), predictors were baseline cognitive flexibility, LNS score and group membership (as defined with three “dummy variables”) and confounders were sex, age and cigarettes smoked per week.

Results

Demographic data and baseline neuropsychological performance

There were no statistically significant differences between the three groups (all p values >0.126) in any demographic or baseline neuropsychological measures (Table 1).

Differences between the three groups in the ID/EDS test

There were significant differences between the groups in the number of total errors (Kruskal-Wallis $\chi^2 = 8.007$, $p = 0.018$), number of completed stages (Kruskal-Wallis $\chi^2 = 7.014$, $p = 0.030$), latency (Kruskal-Wallis $\chi^2 = 8.358$, $p = 0.007$) and total trials (Kruskal-Wallis $\chi^2 = 9.860$, $p = 0.007$). With regard to total errors (Fig. 1A), the fully-trained group outperformed the control group (Mann-Whitney $U = 868.0$, $p = 0.007$, Cohen’s $d = 0.684$) while there was no difference between the control and the partially-trained group (Mann-Whitney $U = 910.5$, $p = 0.056$, Cohen’s $d = 0.429$) as well as between the two experimental groups (Mann-Whitney $U = 853.5$, $p = 0.435$, Cohen’s $d = 0.314$). For the number of completed stages (Fig. 1B), the fully-trained group had superior performance compared with the control group (Mann-Whitney $U = 1080.0$, $p = 0.009$, Cohen’s $d = 0.542$) while no other significant between-group differences were found (control group vs partially-trained group: Mann-Whitney $U = 1096.5$, $p = 0.297$, Cohen’s $d = 0.160$; partially-trained group vs fully-trained group: Mann-Whitney $U = 877.5$, $p = 0.070$, Cohen’s $d = 0.381$). For the measure of latency (Fig. 1C), the control group was slower compared to both

Table 1 Demographic characteristics (Mean $\pm$ SD) and baseline neuropsychological task performance of the three groups

Variables	Control group ($n = 56$)	Partially-Trained group ($n = 42$)	Fully-Trained group ($n = 45$)	p
Age ^a	27.29 $\pm$ 6.58	26.83 $\pm$ 5.46	27.52 $\pm$ 6.6	0.969
Education ^a	17.0 $\pm$ 2.64	16.77 $\pm$ 1.71	16.90 $\pm$ 2.29	0.986
Cigarettes/week ^a	15.5 $\pm$ 37.68	22.64 $\pm$ 43.85	8.27 $\pm$ 25.45	0.126
Raven (raw score) ^a	53.1 $\pm$ 6.04	54.00 $\pm$ 4.51	53.64 $\pm$ 4.63	0.838
Sex (males/females) ^b	33/23	22/20	24/21	0.774
WCST categories completed ^a	5.71 $\pm$ 0.80	5.74 $\pm$ 0.70	5.78 $\pm$ 0.52	0.944
WCST total errors ^a	5.50 $\pm$ 4.79	5.07 $\pm$ 4.36	5.80 $\pm$ 4.07	0.609
WCST unrelated cards ^a	0.96 $\pm$ 2.06	0.81 $\pm$ 1.69	0.64 $\pm$ 1.11	0.555
WCST perseverative errors ^a	1.36 $\pm$ 1.69	1.10 $\pm$ 1.28	1.44 $\pm$ 1.58	0.496
WCST non-perseverative errors ^a	3.16 $\pm$ 2.52	3.17 $\pm$ 2.45	3.71 $\pm$ 2.64	0.550
LNS total score ^a	21.25 $\pm$ 3.33	21.50 $\pm$ 2.89	21.87 $\pm$ 3.06	0.633

LNS Letter-Number Sequencing, WCST Wisconsin Card Sorting test

^aKruskal Wallis test, ^bPearson Chi-square

the partially-trained (Mann-Whitney $U = 812.0$, $p = 0.009$, Cohen's $d = 0.459$) and the fully-trained (Mann-Whitney $U = 932.0$, $p = 0.025$, Cohen's $d = 0.473$) groups, while there was no difference between the two experimental groups (Mann-Whitney $U = 896.0$, $p = 0.677$, Cohen's $d = 0.006$). Finally, the control group required more trials to complete the task (Fig. 1D) compared with both the partially-trained (Mann-Whitney $U = 860.5$, $p = 0.023$, Cohen's $d = 0.434$) and the fully-trained (Mann-Whitney $U = 830.5$, $p = 0.003$, Cohen's $d = 0.591$) groups, again with no difference between the two experimental groups (Mann-Whitney $U = 874.5$, $p = 0.549$, Cohen's $d = 0.159$).

Regression analyses

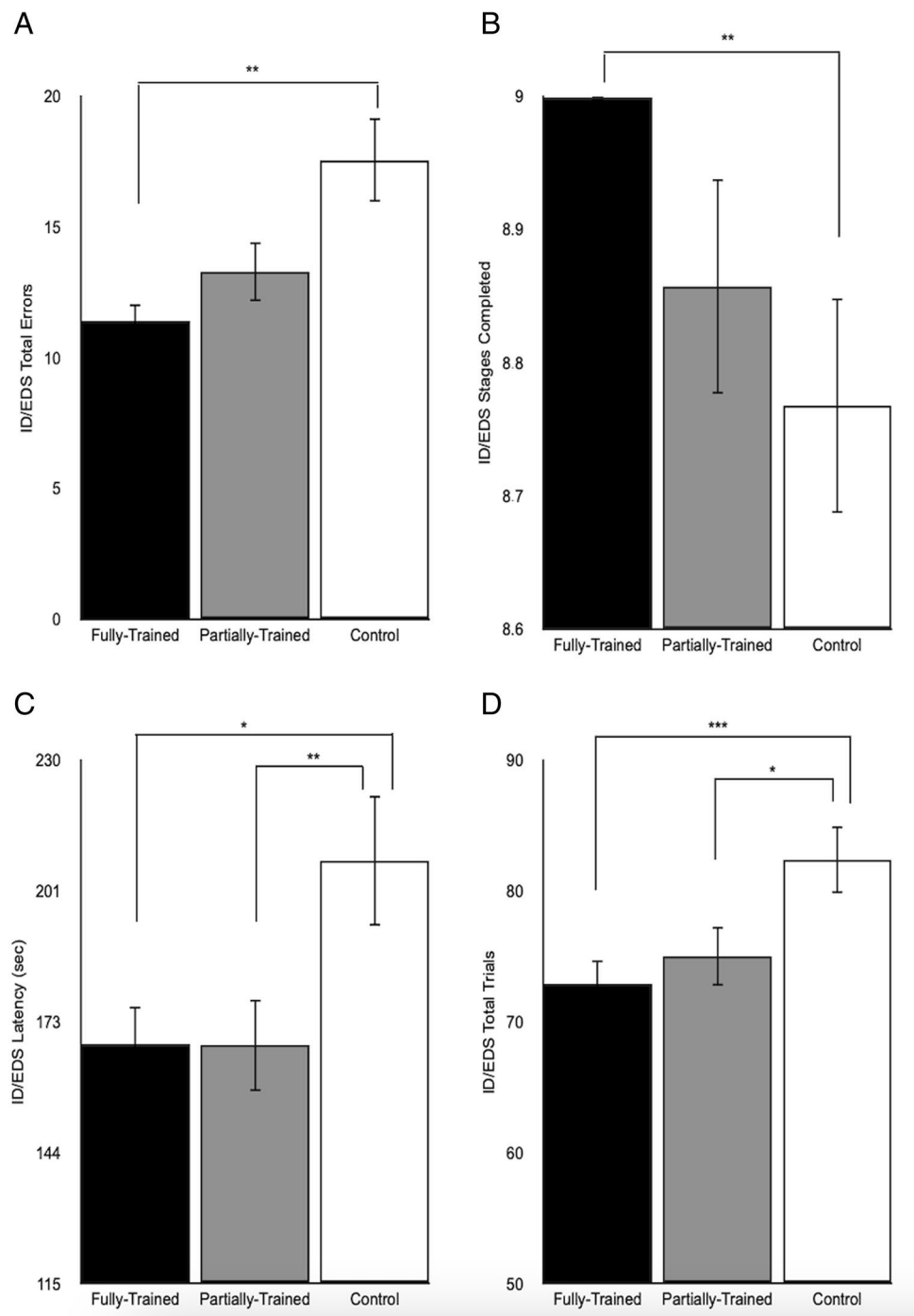
The stepwise regressions revealed significant models for (a) total errors [$F(1,142) = 12.047$, $p < 0.001$, $R^2 = 0.079$] which were associated with inclusion in the control group ($\beta = 0.281$, $t = 3.471$, $p < 0.001$), (b) extra-dimensional shift errors [$F(2,142) = 4.184$, $p = 0.017$, $R^2 = 0.056$] which were associated with poorer baseline cognitive flexibility ($\beta = -0.171$, $t = -2.078$, $p = 0.040$) and inclusion in the control group ($\beta = 0.163$, $t = 1.982$, $p = 0.049$), (c) completed stages [$F(1,142) = 5.219$, $p = 0.024$, $R^2 = 0.036$] which were associated with inclusion in the fully-trained group ($\beta = 0.189$, $t = 2.285$, $p = 0.024$), (d) latency [$F(2,142) = 7.796$, $p < 0.001$, $R^2 = 0.100$] which was associated with inclusion in the control group ($\beta = 0.225$, $t = 2.803$, $p = 0.006$) and male sex ($\beta = -0.210$, $t = -2.610$, $p = 0.010$) and (e) total trials [$F(1,142) = 9.737$, $p = 0.002$, $R^2 = 0.065$] which were associated with inclusion in the control group ($\beta = 0.254$, $t = 3.120$, $p = 0.002$). A detailed description of the results is provided in Supplementary Table 1.

Discussion

Overall, the findings of the study provide preliminary evidence for the existence of far transfer effects of EWM training on cognitive flexibility. In detail, in the between-group analyses, we observed statistically significant differences between the trained and the control groups, as the latter group (a) made more errors and completed fewer stages in comparison with the fully trained group and (b) had increased response latency and required more trials to complete the stages of the cognitive flexibility task compared to both the partially trained and fully trained groups. In the dimensional analyses, the complete lack of EWM training, as evidenced by inclusion in the control group, was associated with more total errors, prolonged response latency and more trials required to complete the cognitive flexibility task as well as with more extradimensional-shift errors (i.e. errors made when participants are required to demonstrate their cognitive flexibility capacities by shifting from one stimulus that was significant up to that point to another one that was not). Baseline cognitive flexibility, though, also contributed to the latter association. In accordance with these, extended short-term EWM training, as evidenced by inclusion in the fully-trained group that received the full training protocol, was associated with more completed stages in the cognitive flexibility task.

As this is the first study on the far transfer effects of EWM training, there is no available literature for direct comparisons. The findings, though, are in agreement with previous studies indicating improvement in aspects of cognitive flexibility after WM training in children (de Vries et al., 2015; Fraser & Cockcroft, 2020), young (Salminen et al., 2012) and older (Chiu et al., 2018) adults. Interestingly, EWM or WM training have been reported to lead to brain-plasticity-related changes (e.g. Miró-Padilla et al., 2020; Miró-Padilla

Fig. 1 Between-group differences in the ID/EDS total errors (**A**), completed stages (**B**), latency (**C**) and total trials (**D**). * $p < 0.05$. ** $p < 0.01$. *** $p < 0.005$



et al., 2021; Salmi et al., 2018; Wu et al., 2021) in the prefrontal-temporal-parietal circuitry that mediates EWM, WM and cognitive flexibility, thus providing an indirect but plausible explanation for the present findings at a neural level. In particular, training seems to “strengthen” this circuitry by either decreasing (Miró-Padilla et al., 2020) or increasing (Salmi et al., 2018) the activation of selected regions or by increasing regional brain volumes (Miró-Padilla et al., 2021) or cortical thickness (Wu et al., 2021) thus leading to

improved performance in the non-trained task. Although we did not find significant differences between the two experimental groups, a linear effect of training is apparent: the fully-trained group had superior performance in all measures followed by the partially-trained and the control groups (see Fig. 1). We, therefore, suggest that the far transfer effects of EWM memory on cognitive flexibility are “dose-dependent”, at least with the design and procedures employed in the present study.

The dimensional analyses revealed an interesting finding: the ability to make shifts at critical stages of the cognitive flexibility task was associated with poorer baseline cognitive flexibility along with receiving training. Previous studies examining the near or far transfer effects of training several cognitive functions [e.g. executive control (Maraver et al., 2020)], have produced analogous results. In this context, our finding indicates that individuals with poorer baseline performance in the non-trained criterion ability (cognitive flexibility in our case) benefit more by the far transfer effects of training another cognitive function (EWM in the present study) and further support the involvement of compensatory means in the process. The compensation effect concept postulates that individuals with lower baseline cognitive abilities benefit more by training as there is space for improvement, while individuals with higher baseline cognitive capacities already demonstrate the optimal level of performance and no further improvement is required and/or is possible (Borella et al., 2017; Maraver et al., 2020).

The majority of studies on cognitive training have thus far focused on children due to implications in academic performance (Gunzenhauser & Nückles, 2021), older adults due to the significant cognitive decline brought about by ageing and its association with daily living (Fan & Wong, 2019), clinical populations (Nousia et al., 2021) or individuals at high-risk for severe psychiatric disorders (e.g. Glenthøj et al., 2017). Contrary to these, the present study involved healthy young adults. Although the study of this population is not directly related to clinical practice, the approach holds certain promise as it provides further information on the processes related to the transfer effects of cognitive training and can serve as a comparison measure for clinical and especially-high risk populations. In addition to this, healthy young adults with poor baseline cognitive flexibility who seem to benefit more by training could also serve as an experimental group for pharmacological studies with cognitive enhancing agents before they are tested in clinical populations.

To conclude, this is the first study providing preliminary evidence for far transfer effects of EWM training on cognitive flexibility. The literature on WM training and its effects is rich but the majority of studies employ tasks (e.g. the n-back task) that rely on effective storage information in an increasing-load manner. In the present study we trained our participants with the more demanding LNS, a task that relies mainly on effective manipulation of stored information prior to recall and is thus more closely associated with cognitive flexibility. It should also be noted that the findings are of significance for cognitive flexibility perceived as an overall latent construct and not for certain aspects of cognitive flexibility (e.g. perseveration, switch cost). This is because we administered different cognitive flexibility tasks at baseline and post-test in order to avoid

practice effects, especially since the 6-day training period was rather short. The lack of administration of the same cognitive flexibility task both at baseline and post-training, though, is a limitation of the present study and should be further considered in future studies administering identical pre- and post-training tasks in order to investigate potential within-subject changes in performance. In addition to this, several factors that might affect the findings still need to be examined in future investigations, as the design of the current study did not permit their inclusion. For example, it is known that genetic factors exert significant influences on the effects of WM training (Bellander et al., 2015), that the motivation of the participants might be a confounding factor (Jaeggi et al., 2014) or that the demands/characteristics of the training task contribute to its transfer effects (Gathercole et al., 2019). Other limitations of the study include the relatively small sample sizes of each group, which might have masked between-group differences, the lack of inclusion of an active control group, and the fact that we did not examine the stability of EWM far transfer effects. The latter is another issue of controversy in the field of cognitive training research (Jaeggi et al., 2014; Nguyen et al., 2019) that is essential to be resolved due to the implications of the findings in clinical practice.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s12144-022-03363-w>.

Acknowledgments The authors would like to thank the participants for their help.

Author Contributions P.B., K.S. and S.G.G. conceptualized and designed the study; V.S. and S.G.G. ran the statistical analyses; V.S. recruited the participants and administered the neuropsychological tasks; K.S. and S.G.G. obtained funding; all authors have contributed in the writing of the manuscript and have approved the submitted version.

Funding The study was partially funded by the Special Accounts for Research of the University of Crete awarded to S.G.G. and K.S. (GA.- No.10214). V.S. was supported by the Hellenic Foundation for Research and Innovation (HFRI) and the General Secretariat for Research and Technology (GSRT), under the HFRI PhD Fellowship grant (GA.- No.4781).

Data Availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Consent to Participate Following oral presentation of the study's aims and methods, all participants received a detailed information sheet and gave written informed consent before inclusion in the study.

Competing Interests The authors declare that there is no conflict of interest.

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